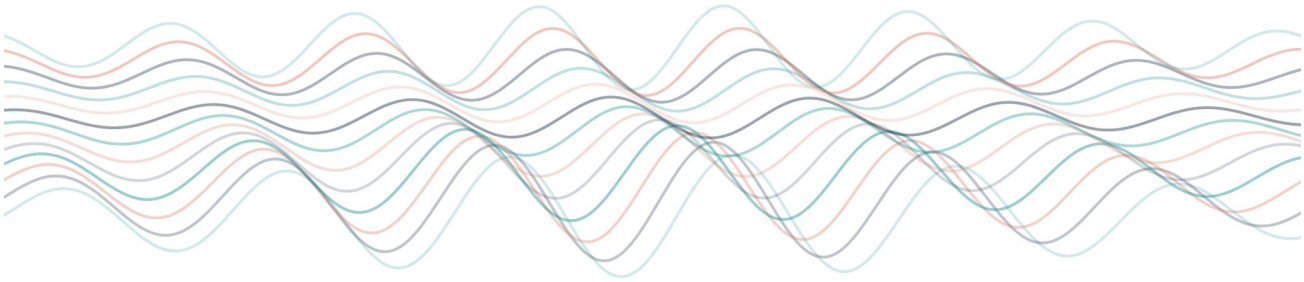
MAHARAH • GRADUATE DEVELOPMENT PROGRAM • AI ASSESSMENT

WeshFahmna AI

CALL CENTER CONVERSATION INTELLIGENCE

Final Technical Project Report



From conversations to clear, traceable insights.

AUTHOR

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PROJECT TYPE

12-call evaluation and visualization prototype

REPORT DATE

3 November 2026

DELIVERY

Offline AI pipeline + deployed Streamlit dashboard

PROJECT LINKS

Live application weshfahmna-ai.streamlit.app

GitHub repository github.com/Yasir-Aladwani/WeshFahmna-AI

01 / EXECUTIVE OVERVIEW

Executive summary

WeshFahmna AI is a research-led prototype that turns role-played, English-language customer-service conversations into structured call summaries, customer sentiment insights, timestamped dialogue, and transparent model-quality assessments. It was developed for the Maharah GDP Call Center Agent Performance Analysis assessment.

Pilot in numbers

CALLS	RECORDED AUDIO	SPEAKER TURNS	CUSTOMER TURNS SCORED
12	118.83 min	776	281

What was delivered

- A research survey comparing 16 candidate datasets and 15 papers/articles; AppTek selected following a documented weighted screening exercise.
- An offline AI workflow covering English ASR, speaker diarization, structured Qwen summaries, and customer sentiment classification.
- Per-call evaluation records, joined CSV/JSON datasets, an audit trail, and a five-view interactive Streamlit interface.
- Documented limitations that distinguish model predictions, reference annotations, and business outcomes not observable in the data.

Headline evaluation results

TASK	12-CALL PILOT RESULT	HOW TO INTERPRET
Speech-to-text	Mean WER 20.85%	Lower is better; arithmetic mean across pilot calls.
Adjusted transcription	Mean WER 18.24%	Filler-handling sensitivity analysis; not standard WER.
Speaker diarization	Mean DER 19.93%	Reference-comparison error, not error-free labeling.
Summarization	Qwen3: 3.83/5; Qwen2.5: 3.31/5	Pilot rubric scores; not generalized benchmarks.
Overall call sentiment	8 neutral; 4 positive	No negative overall calls, despite negative individual turns.

Core conclusion. The project demonstrates an auditable analytics workflow, not a production agent-performance scoring engine. Sentiment, summary quality and speaker-label confidence must be reviewed separately from actual customer satisfaction or resolution.

Evidence basis: project README and integration audit, updated 2 Nov 2026; official AppTek dataset documentation. Report values are documented pilot results, not a newly executed benchmark.

02 / RESEARCH AND DATA SELECTION

Why this dataset?

The assessment began with a structured literature and dataset survey to identify public two-speaker service conversations that support not just transcription, but also segmented roles, summaries and customer-sentiment analysis.

Research decision

RESEARCH ITEM	EVIDENCE / DECISION
Survey coverage	16 candidate datasets and 15 relevant papers/articles.
Highest-ranked dataset	AppTek Call-Center Dialogues - weighted screening score 96/100.
Notable alternative	SpokenWOZ - weighted screening score 85/100.
Interpretation	Screening scores represent project-specific suitability judgments, not externally measured model performance.

Selection criteria

- Task fit: sustained agent-customer interactions rather than unrelated monologues, meetings or scripted short utterances.
- Reference information: verbatim text, role identity, timestamps and speaker identifiers to evaluate ASR and diarization honestly.
- Coverage: several service domains and English accent groups to probe failure modes beyond a single speech style.
- Reuse conditions: public distribution, explicit license and clear evaluation-only restrictions.

Official benchmark versus this pilot

DIMENSION	APPTEK DIARIZATION CONFIG	PROJECT PILOT
Conversations	873	12 sampled calls (seed 42)
Audio format	Merged mono, one file per call	Selected English call recordings
Accent groups	14	4
Service domains	16	8
Timestamped segments	94,679	776 retained turns
Reference roles	Agent / customer	Reference-assisted in app

The AppTek benchmark uses role-played conversations with fictional details, not actual customer support recordings. The official dataset card reports 151.1 hours for the merged diarization configuration; the 12 selected recordings total 118.83 minutes. The larger default configuration reports 128.6 hours of annotated speech; these figures measure different source configurations and should not be conflated.

Data choice. The diarization configuration was selected because it provides single mixed-speaker audio and timestamped agent/customer reference segments suitable for transparent ASR and DER evaluation.

03 / SYSTEM DESIGN

Architecture and processing flow

The prototype separates computationally expensive model execution from a lightweight public dashboard. Experiments were executed offline in Google Colab; Streamlit reads prepared results and does not run speech recognition or large language models when users browse calls.

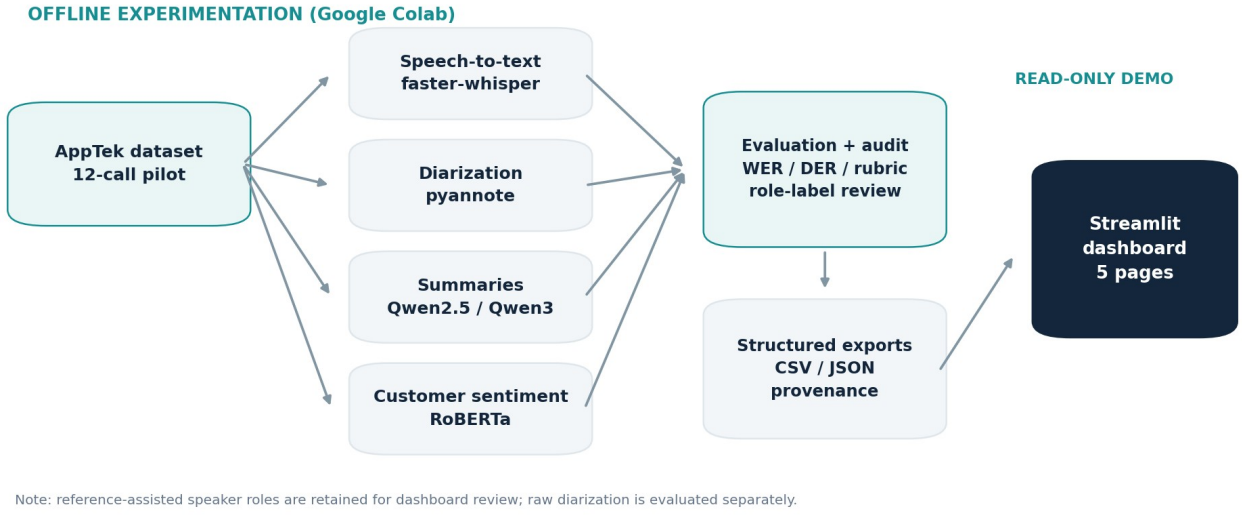


Figure 1. High-level architecture. Reference-assisted role labels support the dashboard, while diarization predictions are assessed as a separate model-quality experiment.

Method at each stage

STEP	IMPLEMENTATION	OUTPUT / CONTROL
1. Select	Metadata-first sampling from AppTek diarization config; seed 42.	12-call pilot; version and source documented.
2. Transcribe	faster-whisper with English small.en model.	Predicted text; WER versus references.
3. Segment	pyannote.audio speaker diarization.	Predicted speaker timing; DER and role audit.
4. Summarize	Qwen2.5-3B and Qwen3-4B-Instruct-2507.	Customer issue; agent action; final outcome; limitations.
5. Classify	Twitter-RoBERTa sentiment classifier.	Customer-turn probabilities and aggregated labels.
6. Integrate	pandas-based validation, exports, dashboard preparation.	Auditable CSV/JSON; Streamlit analytics views.

Important separation. Reference-aligned transcripts and role information were used in downstream review and presentation. Consequently, component-level WER and DER must not be interpreted as a fully independent, end-to-end production accuracy score.

04 / SPEECH ANALYTICS

Transcription and speaker attribution

Quantitative pilot assessment

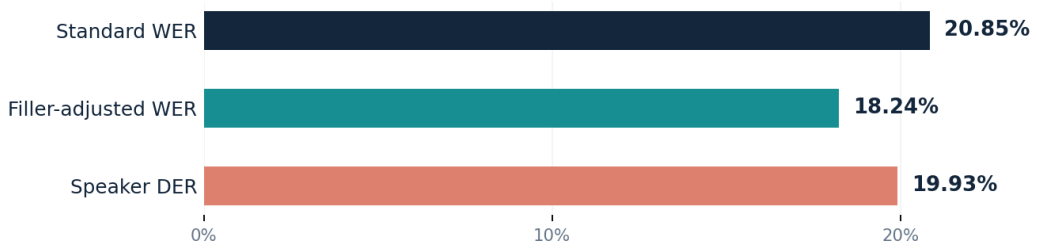


Figure 2. Mean of call-level error percentages across 12 recordings. Lower is better. Original and adjusted WER use different text-processing rules.

MEASURE	DEFINITION	PILOT RESULT
Word Error Rate	(Substitutions + deletions + insertions) / reference words.	20.85% mean
Filler-adjusted WER	WER recalculated after defined filler-token handling.	18.24% mean
Diarization Error Rate	Speaker miss + false alarm + confusion duration relative to reference.	19.93% mean

The adjusted WER is 2.61 percentage points lower than the original reported WER. This is a sensitivity check on transcription normalization, not a demonstrated improvement to the underlying ASR model. Mean percentages are unweighted across the 12 calls, rather than a pooled corpus-wide error calculation.

Failure analysis and safeguards

- CALL_005 showed approximately 55.0% DER and CALL_006 approximately 50.9% DER; both are caution cases for raw speaker attribution.
- The CALL_006 chunking experiment modestly changed ordinary WER from 36.23% to 35.83%, while adjusted WER worsened from 30.69% to 31.20%. The alternative was not clearly better under both measures.
- A total of 32 customer-sentiment records carried an original model speaker tag of Agent while reference-assisted mapping identified them as customer turns.

Reporting rule. The role-labeled Streamlit transcript is reference-assisted. Pyannote DER is reported separately, not hidden behind the reference-corrected display.

05 / LANGUAGE-MODEL EVALUATION

Call summarization and factual quality

Each processed call is condensed into a customer issue, the agent action and the final outcome or next step. Summaries also retain caveats where the transcript cannot verify a claimed action, completion status or commitment.

Qwen comparison

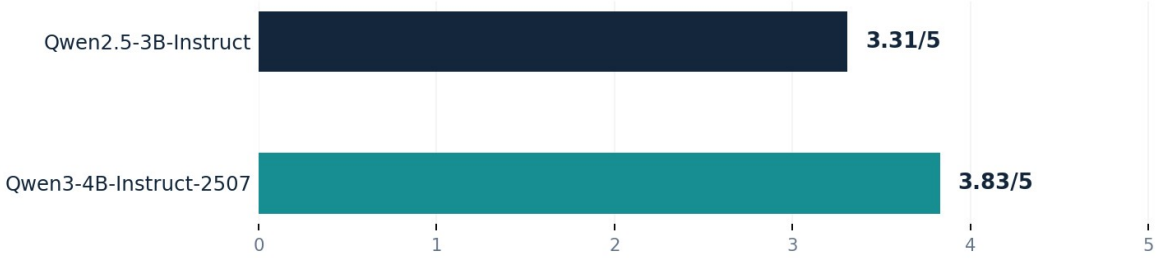


Figure 3. Average pilot rubric score, out of five. The values describe reviewed output quality, not ASR/diarization accuracy.

COMPARISON	RESULT / READING
Qwen2.5-3B-Instruct	Mean overall score: 3.31 / 5
Qwen3-4B-Instruct-2507	Mean overall score: 3.83 / 5
Call-by-call winners	Qwen3 higher in 7; Qwen2.5 higher in 4; tie in 1
Reported paired mean gap	About +0.5 points on the 5-point review scale.

Review rubric and observed shortcomings

- Factual accuracy: Did the summary only state details grounded in the supplied conversation?
- Completeness: Did it cover the customer request, key agent action, and essential constraints?
- Outcome accuracy: Did it distinguish proposals and next steps from actions already completed?
- Lower-scoring Qwen3 calls included CALL_010 (2.00/5) and CALL_012 (2.33/5), supporting case-level caution rather than an unconditional model victory.

The original project audit reports a mean difference of +0.53/5, while subtraction of the two displayed rounded means (3.83 − 3.31) gives +0.52/5. The report therefore states an approximate +0.5-point advantage rather than overstating precision. Raw per-call reviewer scores should be used for any exact comparative claim.

Key lesson. Strong average model performance does not eliminate hallucination or outcome-status errors. Human review remains appropriate before summaries are used for consequential service evaluation.

06 / CUSTOMER-LANGUAGE ANALYSIS

Sentiment patterns and evidence

Two units of analysis are intentionally kept separate: a call-level overall label with justification, and the model-assigned labels and probabilities of 281 selected customer turns. Neither is a ground-truth satisfaction survey.

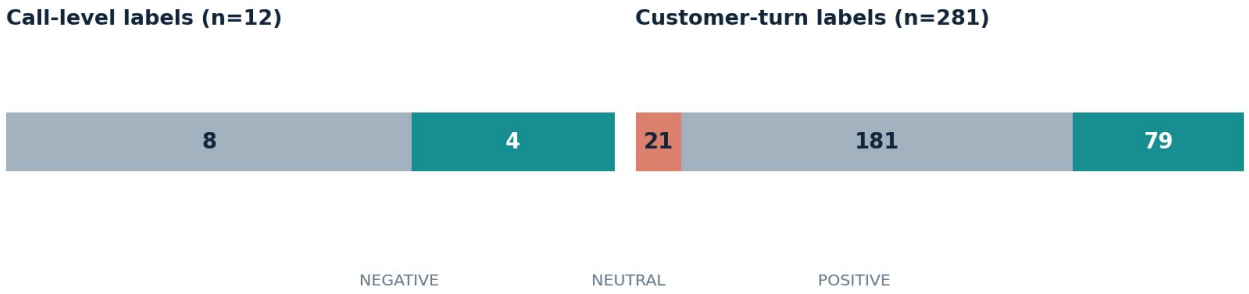


Figure 4. Pilot distributions. Call sentiment and customer-turn sentiment use different denominators and cannot be combined into one rate.

Observed patterns

UNIT / TRANSITION	COUNT	INTERPRETATION
Overall call sentiment	8 neutral; 4 positive; 0 negative	The overall classification is mostly neutral.
Scored customer turns	181 neutral; 79 positive; 21 negative	Negative moments are present despite no negative overall calls.
Neutral → positive	6 calls	Categorical shift, not proof of satisfaction.
Neutral → neutral	3 calls	Stable categorical label across sampled windows.
Positive → neutral	2 calls	Closing language less positive than opening.
Positive → positive	1 call	Positive label at both ends.

What the dashboard adds

- Customer-turn probability outputs, evidence text, and an elapsed-time sentiment timeline for case review.
- A continuous polarity indicator, defined as P(positive) minus P(negative), for exploration only.
- Numeric sentiment-change direction for nine positive-change and three negative-change calls; this is not the same as category transitions.

Interpretation guardrail. The general-purpose Twitter-RoBERTa model is not calibrated here to measure customer satisfaction, escalation, first-contact resolution, or individual employee effectiveness.

07 / PRESENTATION LAYER AND OUTPUTS

Interactive dashboard and reusable data

Five views in the published prototype

DASHBOARD PAGE	WHAT A REVIEWER CAN EXPLORE
Executive Overview	Pilot totals; length, domain and accent distributions; overall sentiment.
Call Explorer	Search and filter calls; issue/action/outcome, timestamped role-labeled transcript, caveats.
Sentiment Insights	Turn-level sentiment probabilities, timelines, opening/closing comparisons.
Model Quality	WER/adjusted WER/DER, per-call outliers, Qwen comparison and quality notes.
About & Methodology	Data provenance, model identities, processing approach and limitations.

Data products retained for audit

ARTIFACT	ROLE
calls.csv + calls.json	Call metadata, summary, headline sentiment, quality metrics, evidence.
speaker_turns.csv	776 timestamped, reference-assisted role-labeled transcript turns.
customer_turn_sentiment.csv	281 sentiment classifications and probability scores.
diarization_metrics.csv	Per-call DER and diarization evaluation statistics.
filler_adjusted_wer.csv	Original versus adjusted speech-recognition error.
model_quality_comparison_12_calls.csv	Paired Qwen rubric quality scores.
speaker_attribution_review.csv	Raw/assisted label discrepancies and attribution review trail.
methodology.json + provenance.json	Model choices, assumptions, data origin and limitations.

Additional final_quality_metrics.csv aggregates the model metrics. Application-ready files can be regenerated using python scripts/prepare_data.py, where that script is included in the checked-out repository. The Streamlit app renders precomputed CSV/JSON views rather than performing live inference.

Direct access

Streamlit demo: https://weshfahmna-ai.streamlit.app/

Source repository: https://github.com/Yasir-Aladwani/WeshFahmna-AI

08 / RELIABILITY, PRIVACY AND COMPLIANCE

Validation and project limitations

Completed project-data consistency checks

- The 12 selected call IDs were aligned across the main call table, transcript table, sentiment file and quality evaluations.
- The original principal CSV files contained no missing required cells in the integration audit; stored sentiment probabilities summed to approximately one per scored turn.
- The transcript contained 7 zero-duration turns. Text was retained, but such rows should not contribute to speaking-time calculations.
- The project prepared a review trail for raw speaker assignment versus dataset-reference-assisted roles, including 32 known role mismatches in sentiment-source records.

Evidence boundaries

CANNOT BE CONCLUDED	WHY NOT
An agent achieved a verified resolution	No ground-truth resolution confirmation or operational case outcome.
A customer was satisfied	No survey-based CSAT or validated satisfaction ground truth.
First-contact resolution / SLA / AHT	No contact-center system instrumentation or agent operational IDs.
Live diarization is accurate	The displayed customer/agent roles were reference-assisted.
Model accuracy generalizes	Only 12 role-played calls; 4 accents and 8 domains sampled.

Responsible data usage

AppTek.ai released the source dataset under CC BY-SA 4.0 for evaluation and analysis. Its published documentation explicitly discourages training/fine-tuning ASR or foundation models on this benchmark. Public materials should attribute AppTek and preserve applicable share-alike requirements when redistributing adapted dataset material. Project code licensing is distinct from dataset licensing.

External-link review. The application and GitHub URLs are the owner-provided project addresses. Availability and anonymous access should be checked in a private/incognito browser before final submission; this report does not certify their current public accessibility.

09 / REPRODUCIBILITY AND NEXT STEPS

Conclusions and recommendations

Reproduce the study - reviewer checklist

A reviewer should clone the source repository, inspect the README and the research survey, install the documented Python requirements, and open the ordered notebooks for dataset inspection, ASR, diarization, summarization, sentiment and final integration. The Streamlit app can be launched locally with streamlit run app.py; the hosted deployment is intended as a read-only exploration of already prepared results.

Full model reproduction requires access to the original AppTek benchmark recordings, compatible model weights and runtime dependencies, and any model-access authorization required by the diarization setup. The report does not claim a fully rerun reproduction audit of all original notebooks.

Prioritized next steps

PRIORITY	ACTION	BENEFIT
1	Recheck report links in incognito and confirm public GitHub contents.	Reliable reviewer access.
2	Expand manually reviewed evaluation beyond 12 calls.	More robust error and accent analysis.
3	Measure end-to-end output quality without assisted labels.	Realistic deployment-risk estimate.
4	Add evidence-linked summary verification and uncertainty flags.	Fewer unsupported outcome claims.
5	Only with consent and appropriate data: validate operational KPIs.	Potential future business usefulness.

Final assessment

Overall result. WeshFahmna AI meets the project's technical objective as a well-documented, 12-call research and visualization prototype. Its distinctive strength is treating error analysis, reference provenance, and limitations as first-class outputs rather than presenting model estimates as verified agent-performance facts.

Sources and project references

[1] AppTek.ai, AppTek Call-Center Dialogues dataset card: huggingface.co/datasets/apptek-com/apptek_callcenter_dialogues

[2] Beck et al. (2026), AppTek Call-Center Dialogues: A Multi-Accent Long-Form Benchmark for English ASR: arxiv.org/abs/2604.27543

[3] Creative Commons Attribution-ShareAlike 4.0: creativecommons.org/licenses/by-sa/4.0

[4] Project repository files: README.md and docs/research_survey.xlsx (accessed 3 Nov 2026); internal integration audit and summary-quality review (2 Nov 2026).

[5] Maharah GDP Trainee Assessment, Call Center Agent Performance Analysis, 2026 (assessment brief).